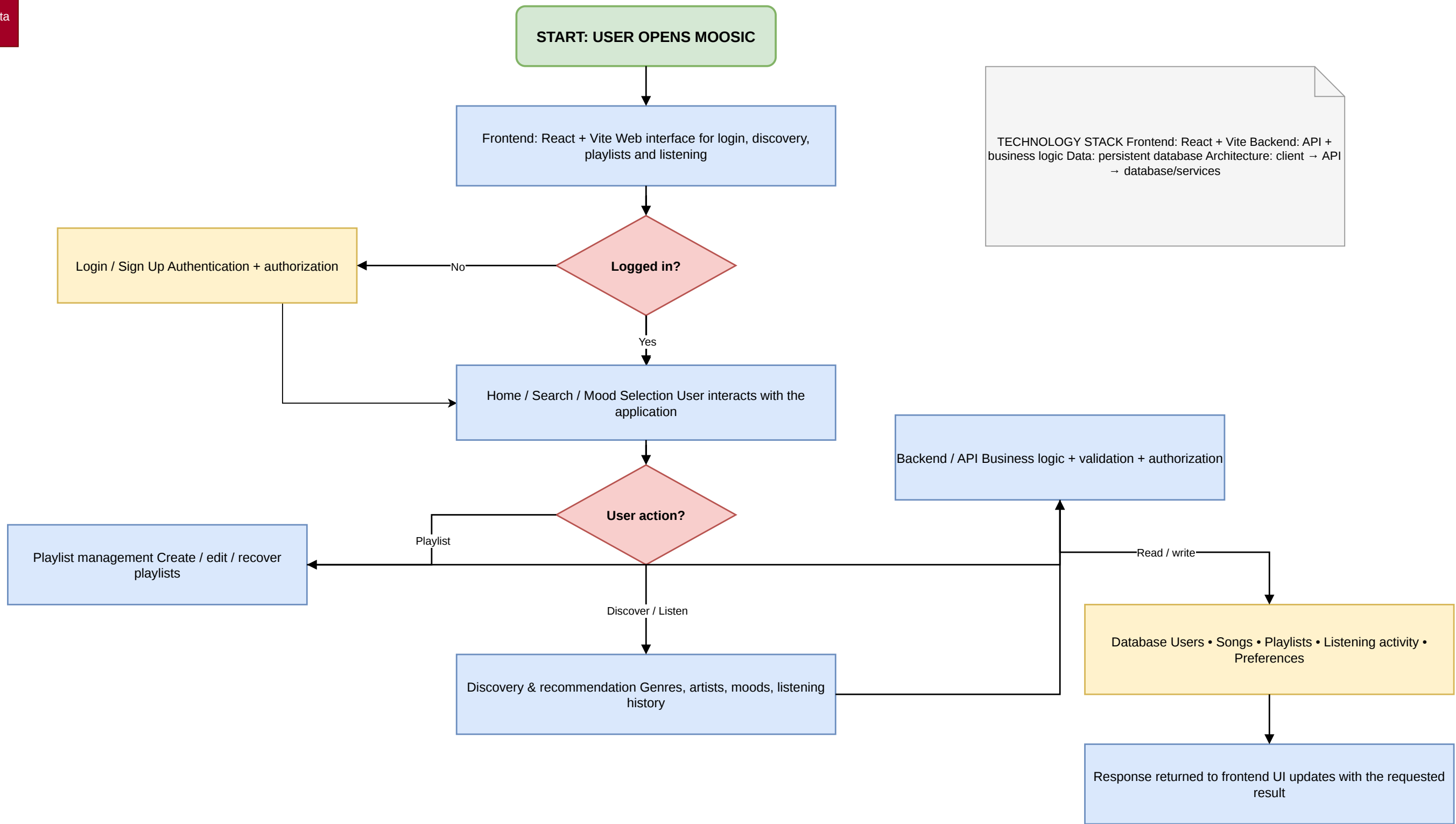
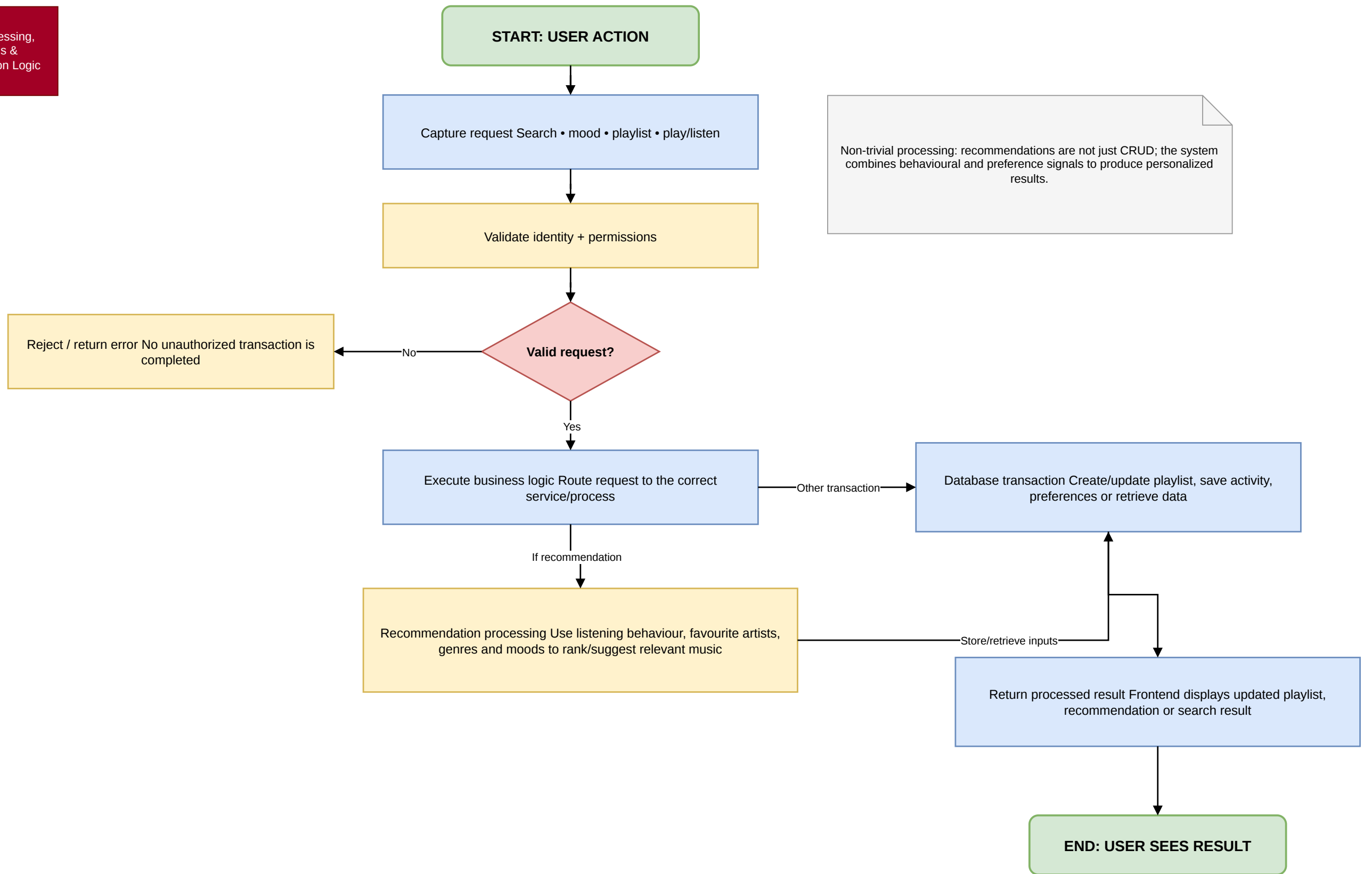
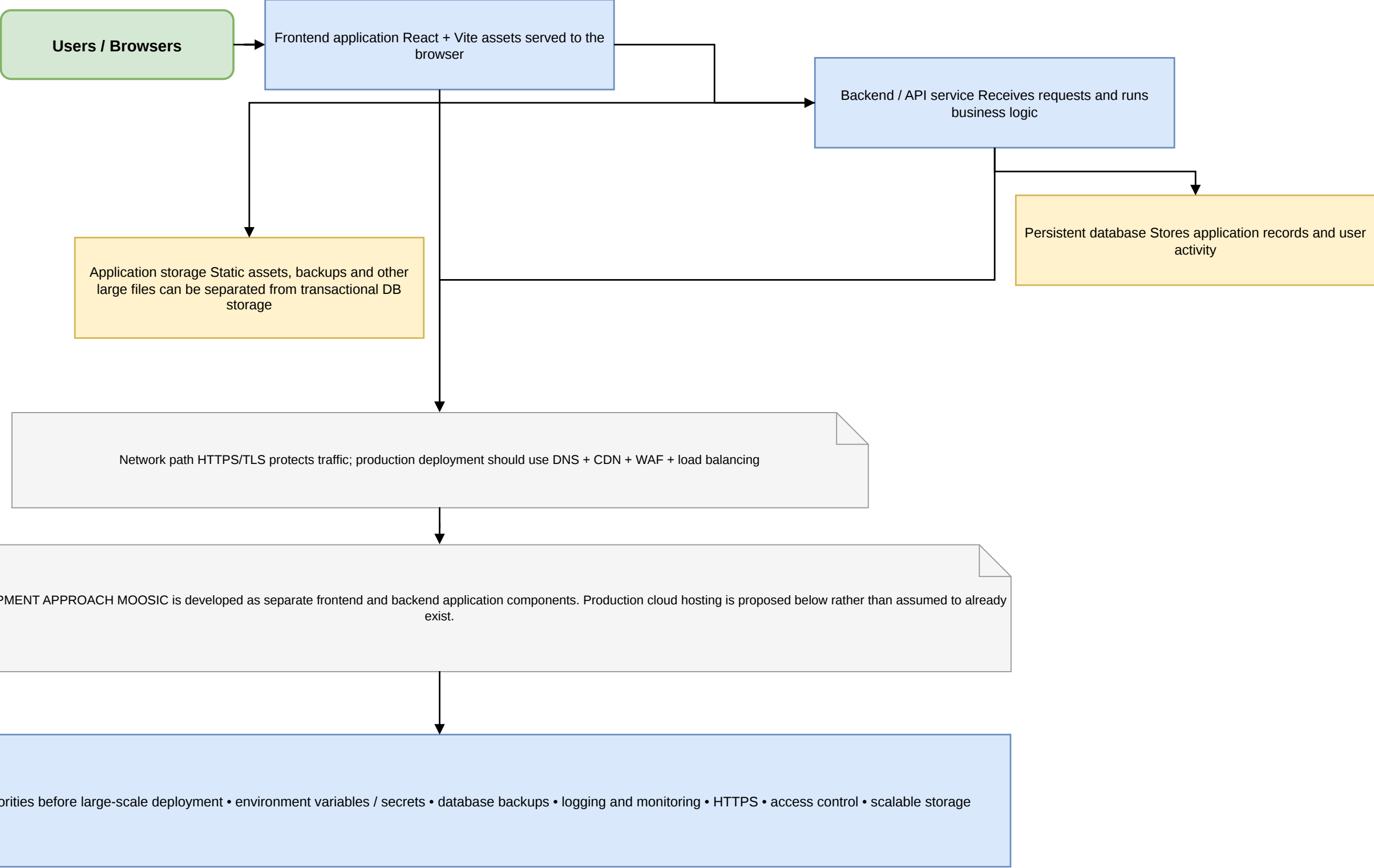
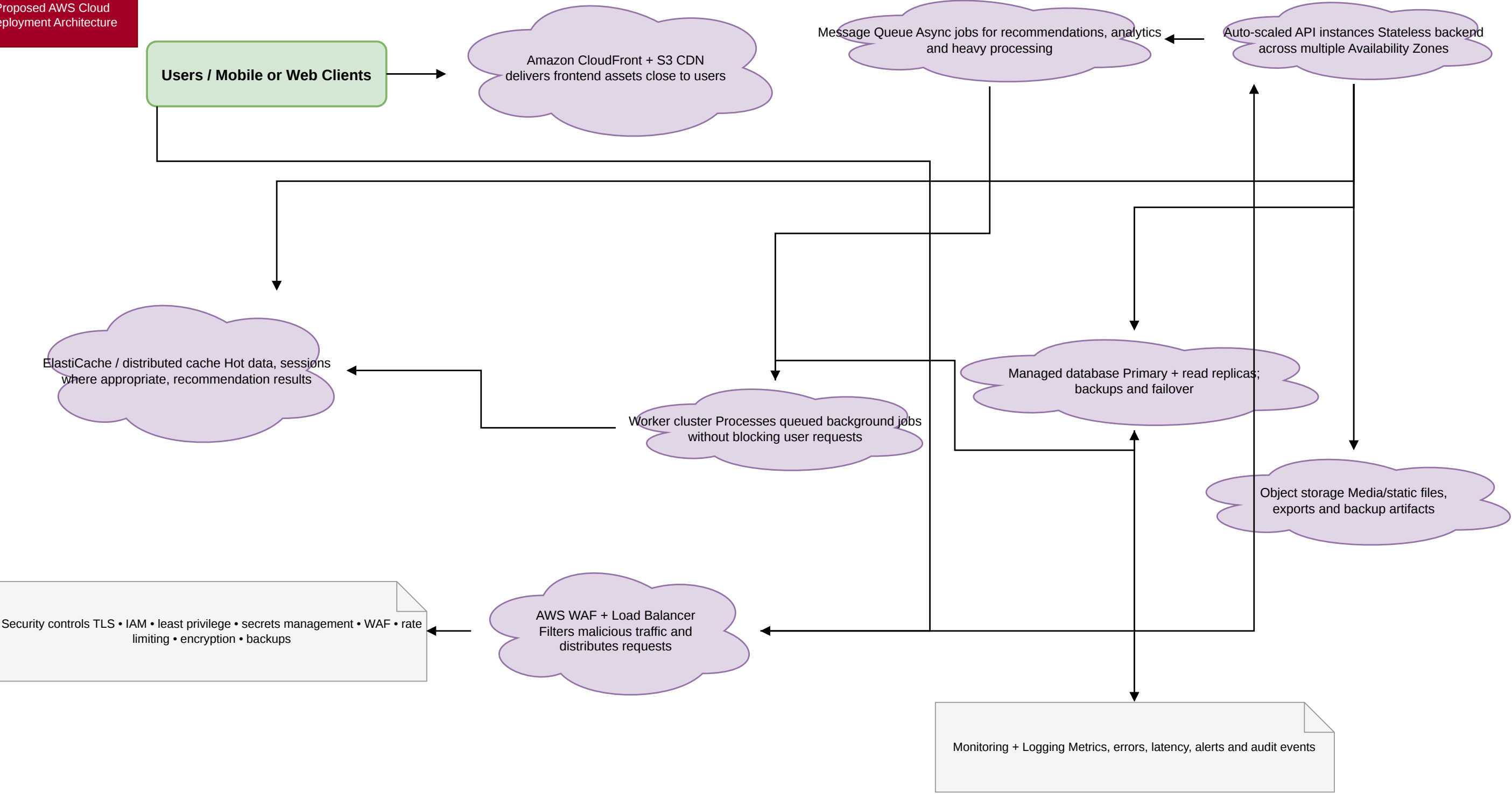


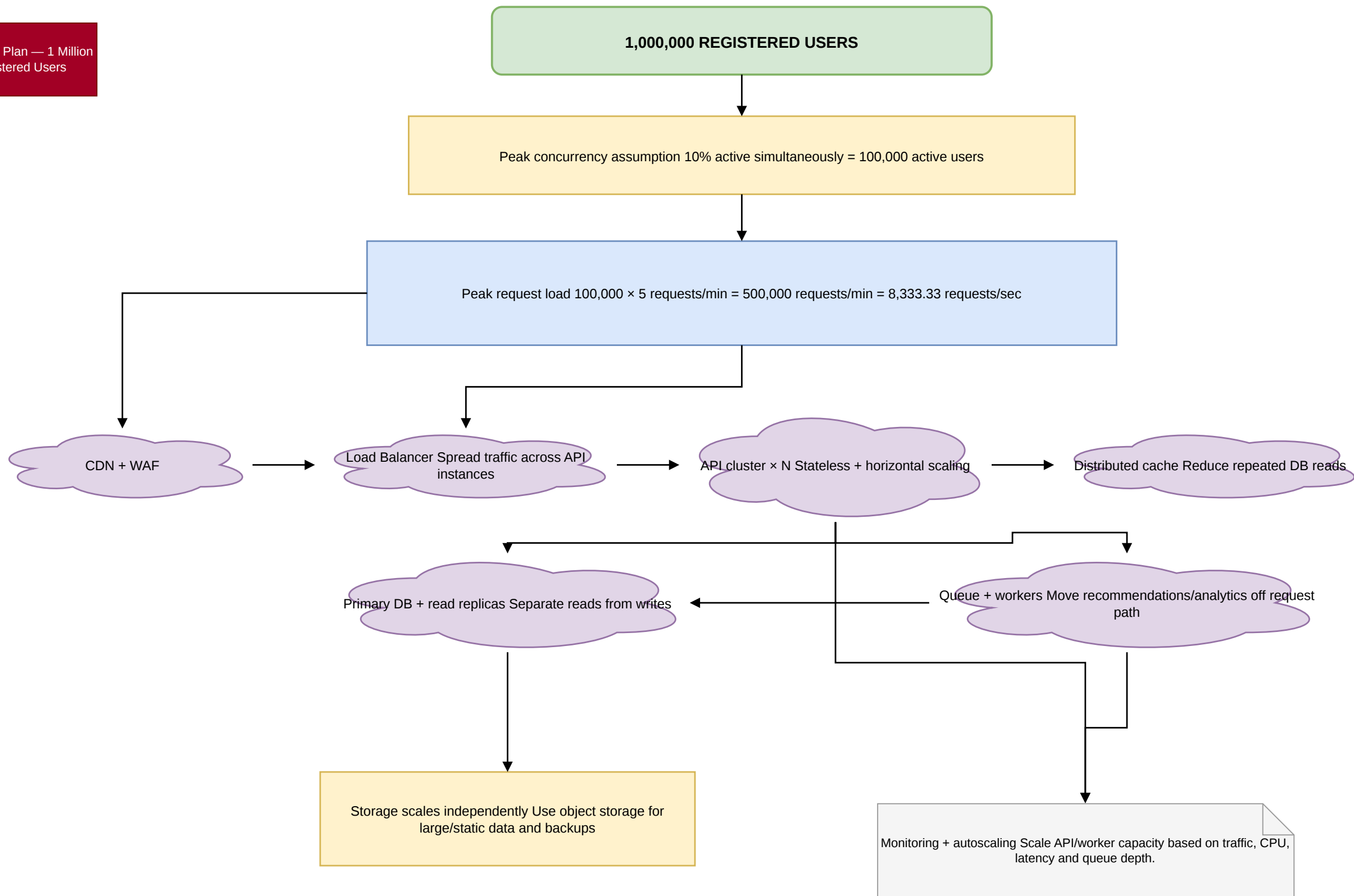






Proposed AWS Cloud
Deployment Architecture





Scalability Plan — 5 Million
Registered Users

5,000,000 REGISTERED USERS

Peak concurrency assumption 10% active simultaneously = 500,000 active users

Peak request load $500,000 \times 5 \text{ requests/min} = 2,500,000 \text{ requests/min} = 41,666.67 \text{ requests/sec}$

Global CDN + WAF Edge delivery +
traffic filtering

Multi-zone / multi-region capable
routing Distribute demand
and isolate failures

Auto-scaled API cluster Many
stateless instances across zones

Distributed cache cluster High-volume
hot reads and recommendation results

Managed DB + read replicas Partition/shard
only if measured data volume requires it

Message queue + worker cluster Async
recommendations, analytics and notifications

Disaster recovery Cross-zone backups, replica strategy, restore testing
and documented RTO/RPO

Continuous monitoring Autoscaling, rate limiting, capacity alerts, error tracking
and performance testing

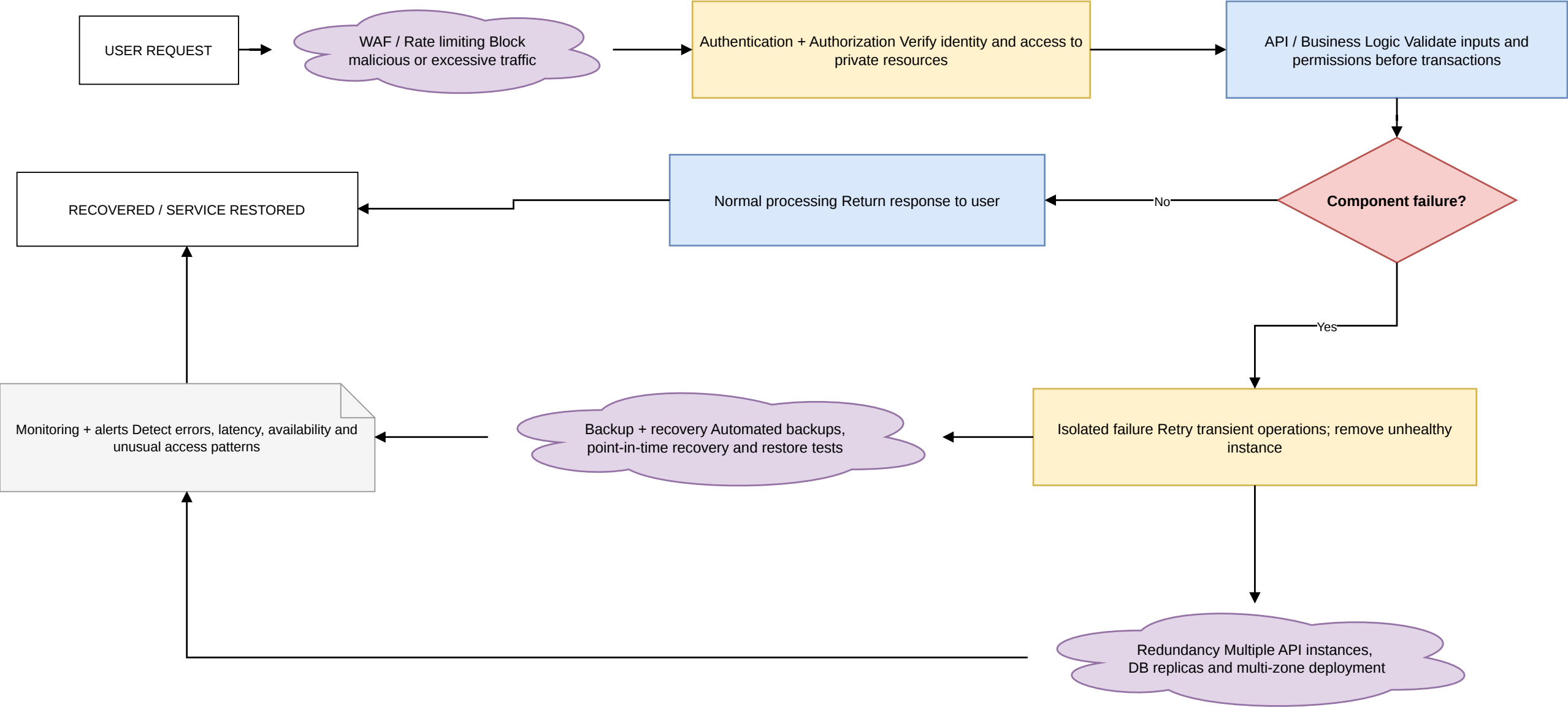
A. USER GROWTH Formula: $\text{Future users} = \text{Initial users} \times (1 + \text{growth rate})^n$
Initial = 10,000; growth = 25% Year 1: 12,500 Year 2: 15,625 Year 3: 19,531 Year 4: 24,414 Year 5: 30,518 Interpretation: registered users increase by 25% each year.

B. PEAK CONCURRENT USERS Formula: $\text{Peak concurrent} = \text{Registered users} \times 10\%$
100,000 → 10,000 active 500,000 → 50,000 active 1,000,000 → 100,000 active 5,000,000 → 500,000 active Interpretation: capacity must support 10% simultaneous activity at peak.

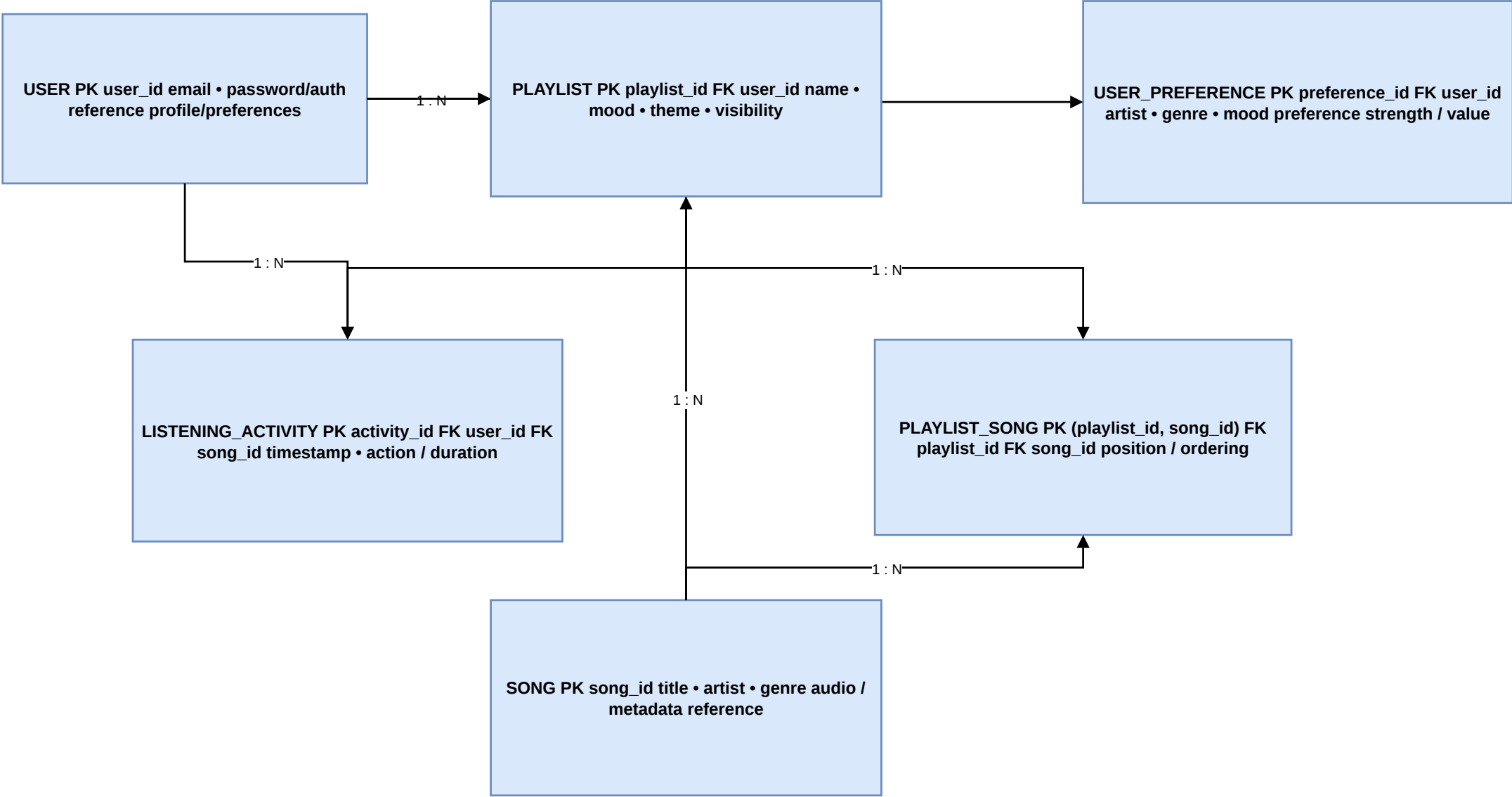
C. REQUESTS PER MINUTE / SECOND Formula: $\text{RPM} = \text{Active users} \times 5 \text{ RPS}$
 $= \text{RPM} \div 60$ 10,000 → 50,000 RPM → 833.33 RPS 50,000 → 250,000 RPM → 4,166.67 RPS 100,000 → 500,000 RPM → 8,333.33 RPS 500,000 → 2,500,000 RPM → 41,666.67 RPS Interpretation: peak traffic grows directly with active users.

Example: 1,000,000 users Formula → Values → Calculation → Result → Interpretation Peak active = $1,000,000 \times 0.10 = 100,000$ Requests/min = $100,000 \times 5 = 500,000$ Requests/sec = $500,000 \div 60 = 8,333.33$ Therefore the architecture should be designed to absorb peak load, not only average traffic.

Security, Failure & Recovery Analysis



Logical ER / Database
Diagram



Logical model based on the documented MOOSIC features. Before final submission, align field names and any additional tables exactly with the implemented backend/database schema.